

Lab 10:

Write a simple streaming program in Spark to receive text data streams on a particular port, perform basic text cleaning (like white space removal, stop words removal, lemmatization, etc.), and print the cleaned text on the screen. (Open Ended Question)

```
import nltk
nltk.download('punkt')
nltk.download('stopwords')
nltk.download('wordnet')

from pyspark.sql import SparkSession
from pyspark.sql.functions import col, lower, regexp_replace, split, explode, udf
from pyspark.sql.types import ArrayType, StringType
from pyspark.ml.feature import StopWordsRemover
from nltk.stem import WordNetLemmatizer

spark = SparkSession.builder.appName("TextProcessing").getOrCreate()

lines = [
    "Hello, I hate you.",
    "I hate that I love you.",
    "Don't want to, but I can't put",
    "nobody else above you."
]

df = spark.createDataFrame(lines, "string").toDF("value")

# Step 1: Lowercase + remove punctuation
df_clean = df.select(
    regexp_replace(lower(col("value")), "[^a-zA-Z\s]", "").alias("cleaned")
)

# Step 2: Tokenize
df_tokens = df_clean.select(split(col("cleaned"), "\s+").alias("tokens"))

# Step 3: Remove stop words
remover = StopWordsRemover(inputCol="tokens", outputCol="filtered")
df_filtered = remover.transform(df_tokens)

# Step 4: Lemmatization
lemmatizer = WordNetLemmatizer()
def lemmatize_words(words):
    return [lemmatizer.lemmatize(word) for word in words]
lemmatize_udf = udf(lemmatize_words, ArrayType(StringType()))
df_lemmatized = df_filtered.withColumn("lemmatized", lemmatize_udf(col("filtered")))

# Step 5: Show result
df_lemmatized.select(explode(col("lemmatized")).alias("word")).show(truncate=False)

Requirement already satisfied: nltk in /usr/local/lib/python3.11/dist-packages (3.9.1)
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data]   Unzipping tokenizers/punkt.zip.
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data]   Unzipping corpora/stopwords.zip.
```

[nltk_data] Downloading package wordnet to /root/nltk_data...
2026-05-03 15:22:41 INFO SparkSession: Creating Spark session

+-----+	
word	
+-----+	
hello	
hate	
hate	
love	
dont	
want	
cant	
put	
nobody	
else	
+-----+	